



Qualification Pack

Vermimanuring Technician

QP Code: CUTM/AGR/Q0901

Version: 1.0

NSQF Level: 4

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CUTM/AGR/Q0901: Vermimanuring Technician

Brief Job Description

The Individual is responsible to understand the vermimanuring and production of vermimanuring, selection of composting material (organic and inorganic) and worm species, vermibed preparation and value addition of vermicomposting and by-products for the sustainable agriculture production.

Personal Attributes

A Vermimanuring Technician should be environmentally conscious, physically fit, and attentive to detail. Patience, cleanliness, and responsibility are essential for managing earthworms and organic waste. The individual should possess basic technical and observational skills, a willingness to work outdoors, and a positive attitude toward sustainable farming and soil health improvement.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/AGR/N0901: Handling of Vermimanuring](#)
2. [CUTM/AGR/N0902: The Role of Earthworms in Vermimanuring](#)
3. [CUTM/AGR/N0903: Choosing the Right Earthworm Species](#)
4. [CUTM/AGR/N0904: Advanced Vermimanuring Techniques](#)
5. [CUTM/AGR/N0905: Value added to Vermimanuring and Preservation](#)
6. [CUTM/AGR/N0906: Quality Assurance in Vermimanuring](#)
7. [CUTM/AGR/N0907: Operation And Biowaste Management](#)
8. [CUTM/VSQ/N7803: Maintain Health and Safety at Work Place.](#)
9. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Sustainable Agriculture
Occupation	Animal Producers and Related Workers, Other



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Country	India
NSQF Level	4
Credits	19
Aligned to NCO/ISCO/ISIC Code	NSQF/NCrF
Minimum Educational Qualification & Experience	12th grade Pass with NA of experience OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	21/10/2027
NSQC Approval Date	22/10/2024
Version	1.0
Reference code on NQR	QG-04-AG-03355-2024-V1-CUTM
NQR Version	1.0

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CUTM/AGR/N0901: Handling of Vermimanuring

Description

This NOS focuses on the foundational steps of handling vermimanuring, including compost material selection, vermibed preparation, inoculation of worms, and maintenance of composting environments to produce high-quality organic manure.

Scope

The scope covers the following :

- Involves hands-on activities for setting up vermibeds, monitoring physico-chemical conditions, and post-processing vermicompost. Learners gain skills in initiating and managing vermimanuring operations effectively for sustainable agriculture.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the best method of preparing vermibed and selection of suitable species.
- PC2.** Selection of composting materials.
- PC3.** Pre-processing techniques of vermimanuring and inoculation of earthworms.
- PC4.** Analysis of chemical and physical parameters during vermimanuring.
- PC5.** Harvest and post harvesting process of vermimanure and it's by- Product.
- PC6.** Identify different equipment for microbial and chemical analysis.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Concepts and methods of preparing vermibeds.
- KU2.** Types of compostable organic materials.
- KU3.** Pre-processing techniques for compost.
- KU4.** Physico-chemical parameters in composting (moisture, pH, aeration).
- KU5.** Earthworm inoculation practices. Post-harvest handling of vermicompost and by-products.
- KU6.** Safety and sanitation protocols during handling.
- KU7.** Tools and equipment for vermibed preparation.
- KU8.** Importance of temperature, humidity, and water activity.
- KU9.** Monitoring vermibed health and quality parameters.

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** Prepare composting material and vermibeds.
- GS2.** Operate and maintain composting setups.
- GS3.** Analyze compost quality through basic testing tools.
- GS4.** Identify and address issues in the composting process.
- GS5.** Follow sequence of operations in vermimanuring.
- GS6.** Handle worms without causing stress or loss.
- GS7.** Maintain workflow consistency and recordkeeping.
- GS8.** Communicate composting methods and results effectively.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. Identify the best method of preparing vermicompost and selection of suitable species.	4	5	-	5
PC2. Selection of composting materials.	6	4	-	5
PC3. Pre-processing techniques of vermicomposting and inoculation of earthworms.	5	9	-	5
PC4. Analysis of chemical and physical parameters during vermicomposting.	5	9	-	5
PC5. Harvest and post harvesting process of vermicompost and its by- Product.	5	6	-	5
PC6. Identify different equipment for microbial and chemical analysis.	5	7	-	5
NOS Total	30	40	-	30



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0901
NOS Name	Handling of Vermimanuring
Sector	Agriculture
Sub-Sector	
Occupation	Animal Producers and Related Workers, Other
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024

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CUTM/AGR/N0902: The Role of Earthworms in Vermimanuring

Description

This NOS explains the ecological, biological, and economic roles of earthworms in vermimanuring. Learners understand how worms convert organic waste into rich manure and improve soil fertility and sustainability.

Scope

The scope covers the following :

- Focuses on recognizing worm types, their contributions to soil health, and their effective role in the vermicomposting ecosystem. It prepares learners to manage and optimize earthworm-based organic recycling.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Types of earthworms used in Vermimanuring.
- PC2.** Anatomy and life cycle of earthworms.
- PC3.** Earthworm ecology and their role in soil health.
- PC4.** Differences and similarities of earthworms.
- PC5.** Setting Up a Vermimanuring System.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Earthworm anatomy and life cycle.
- KU2.** Ecology and behavioral traits of composting worms.
- KU3.** Role in soil structure, aeration, and microbial enrichment.
- KU4.** Differences among composting and burrowing worms.
- KU5.** Earthworm activity in nutrient cycling.
- KU6.** Contribution to sustainable agriculture.
- KU7.** Waste decomposition process via earthworms.
- KU8.** Organic matter transformation to vermicompost.
- KU9.** Environmental and economic impact.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Identify key composting earthworm species.
- GS2.** Set up and monitor worm activity in compost.



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- GS3.** Assess worm population growth and health.
- GS4.** Track improvement in soil structure post-application.
- GS5.** Evaluate worm-based waste conversion efficiency.
- GS6.** Maintain worm beds and monitor feed.
- GS7.** Handle worms without harming their lifecycle.
- GS8.** Promote environmental sustainability through vermicomposting.
- GS9.** Record worm productivity data.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	30	-	-
PC1. Types of earthworms used in Vermimanuring.	4	6	-	-
PC2. Anatomy and life cycle of earthworms.	4	6	-	-
PC3. Earthworm ecology and their role in soil health.	4	6	-	-
PC4. Differences and similarities of earthworms.	4	6	-	-
PC5. Setting Up a Vermimanuring System.	4	6	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0902
NOS Name	The Role of Earthworms in Vermimanuring
Sector	Agriculture
Sub-Sector	
Occupation	Animal Producers and Related Workers, Other
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024

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CUTM/AGR/N0903: Choosing the Right Earthworm Species

Description

This NOS guides the learner on selecting earthworm species best suited for vermicomposting. It enables efficient organic waste decomposition and production of nutrient-rich compost by aligning environmental and species-specific factors.

Scope

The scope covers the following :

- Focuses on species identification, suitability to climatic and feed conditions, and management practices. Learners are trained to choose and manage worms to optimize vermicomposting results.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Eisenia fetida (Red Wiggler)
- PC2.** Eudrilus eugeniae (African Nightcrawler)
- PC3.** Perionyx excavatus (Indian Blue)
- PC4.** Building Types of bins (plastic, wooden, stacked, continuous flow)
- PC5.** Types of feedstock (fruit and vegetable scraps, coffee grounds, eggshells)
- PC6.** Feeding schedules and quantities
- PC7.** Identifying and solving problems (odor, pests, slow composting rates)
- PC8.** Health indicators of a thriving Vermicomposting system

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Species identification (e.g., Eisenia fetida, Lumbricus rubellus).
- KU2.** Environmental suitability of different species.
- KU3.** Decomposition capacity of earthworm types.
- KU4.** Resilience to environmental stressors.
- KU5.** Earthworm adaptability to feedstocks.
- KU6.** Disease resistance among species.
- KU7.** Role in enhancing soil nutrients.
- KU8.** Impact on composting efficiency.
- KU9.** Species-specific maintenance needs.
- KU10.** Benefits for large-scale composting.

Generic Skills (GS)



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User/individual on the job needs to know how to:

- GS1.** Identify and distinguish composting species.
- GS2.** Match worms to specific waste types.
- GS3.** Adjust compost environments to species preferences.
- GS4.** Monitor worm behavior and output.
- GS5.** Evaluate species for climatic suitability.
- GS6.** Troubleshoot species-related issues.
- GS7.** Implement species-specific care routines.
- GS8.** Enhance composting through optimal species use.
- GS9.** Record species performance in different setups.
- GS10.** Train others on worm selection.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	30	-	-
PC1. Eisenia fetida (Red Wiggler)	1	4	-	-
PC2. Eudrilus eugeniae (African Nightcrawler)	2	4	-	-
PC3. Perionyx excavatus (Indian Blue)	2	4	-	-
PC4. Building Types of bins (plastic, wooden, stacked, continuous flow)	2	5	-	-
PC5. Types of feedstock (fruit and vegetable scraps, coffee grounds, eggshells)	2	5	-	-
PC6. Feeding schedules and quantities	3	2	-	-
PC7. Identifying and solving problems (odor, pests, slow composting rates)	4	2	-	-
PC8. Health indicators of a thriving Vermicomposting system	4	4	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0903
NOS Name	Choosing the Right Earthworm Species
Sector	Agriculture
Sub-Sector	
Occupation	Animal Producers and Related Workers, Other
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024

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CUTM/AGR/N0904: Advanced Vermicomposting Techniques

Description

This NOS enables learners to perform vermicomposting at an advanced level using scientific methods and tools. It includes process optimization, quality management, and troubleshooting in large-scale operations.

Scope

The scope covers the following :

- Covers improved systems, inputs, and operational methods to scale vermicomposting efficiently. Learners acquire skills for advanced handling and management practices suited for commercial organic farming and waste recycling.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Continuous flow systems.
- PC2.** Multi-tiered systems.
- PC3.** Integration with other sustainable practices (rainwater harvesting, permaculture)
- PC4.** Adjusting practices for various climates (tropical, temperate, arid)
- PC5.** Indoor versus outdoor systems.
- PC6.** Role of Vermicomposting in waste reduction.
- PC7.** Educational outreach and community projects.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Advanced layering and bin-based composting systems.
- KU2.** Temperature and moisture control strategies.
- KU3.** Accelerated composting using microbial inoculants.
- KU4.** Troubleshooting slow or incomplete composting.
- KU5.** Management of large-scale vermicomposting units.
- KU6.** Prevention and control of pathogens and pests.
- KU7.** Integration with other sustainable farming practices.
- KU8.** Nutrient optimization through input adjustment.
- KU9.** Pre-treatment of complex bio-wastes.

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** Operate modular and stacked vermibed systems.
- GS2.** Apply compost accelerators or microbial cultures.
- GS3.** Monitor system conditions using instruments.
- GS4.** Perform preventive maintenance of units.
- GS5.** Manage workflow in medium/large-scale operations.
- GS6.** Rectify issues like foul odor, compaction, or pest infestation.
- GS7.** Implement SOPs for consistent compost output.
- GS8.** Document process variations and outcomes.
- GS9.** Optimize feedstock mix for yield improvement.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. Continuous flow systems.	4	10	-	-
PC2. Multi-tiered systems.	5	10	-	-
PC3. Integration with other sustainable practices (rainwater harvesting, permaculture)	5	10	-	-
PC4. Adjusting practices for various climates (tropical, temperate, arid)	5	10	-	-
PC5. Indoor versus outdoor systems.	5	10	-	-
PC6. Role of Vermicomposting in waste reduction.	3	10	-	-
PC7. Educational outreach and community projects.	3	10	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0904
NOS Name	Advanced Vermimanuring Techniques
Sector	Agriculture
Sub-Sector	
Occupation	Animal Producers and Related Workers, Other
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024

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CUTM/AGR/N0905: Value added to Vermimanuring and Preservation

Description

This NOS involves enhancing the value of vermicompost through improved quality, preservation, and packaging practices. It covers post-production handling to prepare the product for market or long-term use.

Scope

The scope covers the following :

- Focuses on sieving, enrichment, vermiwash collection, packaging, storage, and product grading. Learners gain essential skills to prepare compost for commercial sale and ensure customer satisfaction.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Knowledge about various products produced from vermicomposting for agricultural aspects.
- PC2.** Vermicompost is a quality source for improve soil nutrient and fertility.
- PC3.** Vermiwash/Vermifilters as a byproduct aid to improve soil fertility as well as function as a pesticides.
- PC4.** Vermiculture is a preserving technique for mass production of various beneficiary earthworm species for recycling of vermibed formation.
- PC5.** Understand the end product quality for preservation.
- PC6.** Knowledge on natural preservative applicable of .vermicompost on green revolution.
- PC7.** Understanding on packaging and storage methods of value-added vermicomposting products.
- PC8.** Understand different parameters involved in the preservation (time, temperature, humidity, water activity)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Techniques to improve nutrient density in vermicompost.
- KU2.** Sieving, drying, and granulation processes.
- KU3.** Vermiwash collection and usage.
- KU4.** Packaging materials and labeling standards.
- KU5.** Storage conditions to maintain quality.
- KU6.** Shelf-life and contamination risks.
- KU7.** Market grading and pricing standards.
- KU8.** Branding and local value addition practices.
- KU9.** Regulatory norms for compost marketing.



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KU10. Safe transportation of organic products.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Process vermicompost for uniform quality.
- GS2.** Use drying and packaging tools effectively.
- GS3.** Collect and preserve vermiwash hygienically.
- GS4.** Label and store compost as per standards.
- GS5.** Grade products based on quality parameters.
- GS6.** Maintain inventory and sales records.
- GS7.** Coordinate with packaging suppliers and retailers.
- GS8.** Prevent product spoilage and contamination.
- GS9.** Respond to customer quality queries.
- GS10.** Demonstrate value addition to clients and buyers.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Knowledge about various products produced from vermimanuring for agricultural aspects.	5	-	-	-
PC2. Vermimanure is a quality source for improve soil nutrient and fertility.	5	10	-	-
PC3. Vermiwash/Vermifilters as a byproduct aid to improve soil fertility as well as function as a pesticides.	5	10	-	-
PC4. Vermiculture is a preserving technique for mass production of various beneficiary earthworm species for recycling of vermibed formation.	5	10	-	-
PC5. Understand the end product quality for preservation.	5	10	-	-
PC6. Knowledge on natural preservative applicable of .vermimanure on green revolution.	5	-	-	-
PC7. Understanding on packaging and storage methods of value-added vermimanuring products.	5	10	-	-
PC8. Understand different parameters involved in the preservation (time, temperature, humidity, water activity)	5	10	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0905
NOS Name	Value added to Vermimanuring and Preservation
Sector	Agriculture
Sub-Sector	
Occupation	Animal Producers and Related Workers, Other
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024

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CUTM/AGR/N0906: Quality Assurance in Vermimanuring

Description

This NOS focuses on maintaining high standards of quality in vermimanuring through regular testing and documentation. It ensures that compost meets market, environmental, and organic certification standards.

Scope

The scope covers the following :

- Covers quality control checks, internal audits, documentation, and sample testing. Learners gain skills to ensure product reliability and compliance with certification norms, enhancing consumer trust.

Elements and Performance Criteria

Understanding the different quality parameters associated with process of Vermimanuring

To be competent, the user/individual on the job must be able to:

- PC1.** Maintaining and managing of important quality parameters to obtain standard compost within prescribed limits.
- PC2.** Understand the activity of worms and microbes for nutrient content in vermimanure.

Knowledge on various methods of assessment and maintenance of quality

To be competent, the user/individual on the job must be able to:

- PC3.** Harvesting and post-harvest technology to maintain the quality of product and by-product.
- PC4.** Biochemical, microbiological equipment and methods used for quality control measure.
- PC5.** Testing and analysis for factors affecting the quality of vermimanure.
- PC6.** Measurement of Hazard Analysis and Critical Control Point (HACCP)
- PC7.** Quality certification standards.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Testing protocols and equipment.
- KU2.** Documentation for traceability.
- KU3.** Quality benchmarks for organic compost.
- KU4.** Common contamination issues.
- KU5.** Sampling procedures and batch coding.
- KU6.** Corrective action methods.
- KU7.** Internal auditing practices.
- KU8.** Certification and labeling requirements.
- KU9.** Consumer feedback handling.



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Perform basic tests for pH, moisture, etc.
- GS2.** Collect and label samples for lab testing.
- GS3.** Maintain logs and quality records.
- GS4.** Identify and separate substandard batches.
- GS5.** Coordinate internal checks and audits.
- GS6.** Document corrective actions.
- GS7.** Communicate non-compliance to stakeholders.
- GS8.** Respond to quality-related inquiries.
- GS9.** Implement quality improvements based on feedback.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding the different quality parameters associated with process of Vermimanuring</i>	11	10	-	-
PC1. Maintaining and managing of important quality parameters to obtain standard compost within prescribed limits.	5	10	-	-
PC2. Understand the activity of worms and microbes for nutrient content in vermicompost.	6	-	-	-
<i>Knowledge on various methods of assessment and maintenance of quality</i>	29	50	-	-
PC3. Harvesting and post-harvest technology to maintain the quality of product and by-product.	5	10	-	-
PC4. Biochemical, microbiological equipment and methods used for quality control measure.	7	10	-	-
PC5. Testing and analysis for factors affecting the quality of vermicompost.	5	10	-	-
PC6. Measurement of Hazard Analysis and Critical Control Point (HACCP)	8	10	-	-
PC7. Quality certification standards.	4	10	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0906
NOS Name	Quality Assurance in Vermimanuring
Sector	Agriculture
Sub-Sector	
Occupation	Animal Producers and Related Workers, Other
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024

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CUTM/AGR/N0907: Operation And Biowaste Management

Description

This NOS outlines how to manage organic waste effectively for vermicomposting. It includes segregation, processing, and safe handling of bio-waste while ensuring environmental sustainability and compliance with safety norms.

Scope

The scope covers the following :

- Includes identifying, collecting, and processing bio-waste safely for use in composting. Learners gain skills to manage waste streams and apply hygienic practices in a sustainable manner within vermicomposting operations.

Elements and Performance Criteria

Information about handling of waste

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the composition of compost materials and vermicompost towards proper biowaste management.
- PC2.** Understand the difference between vermicomposting and vermiculture for handling biowaste materials.
- PC3.** Clearly understand the role of worms along with microbes used for conversion of Biobiowaste during vermicomposting.

Knowledge on productivity and sustainable practice

To be competent, the user/individual on the job must be able to:

- PC4.** Understand detailed information about suitable biowastes for vermicomposting with proper providing parameters to obtain quality product for enhancing soil nutrition.
- PC5.** Knowledge on different sustainable processing practices for agriculture approaches.
- PC6.** Information about the different kinds of by-products and utilization in innovative agronomic process.
- PC7.** Know about best farming practices by integrating vermicompost for sustainable environmental approaches.

Knowledge on production of high value end product

To be competent, the user/individual on the job must be able to:

- PC8.** Know the biochemistry and microbial aspects of different earthworm.
- PC9.** Selection and application of appropriate method for production of high value product.
- PC10.** Biochemical analysis of high value end products.
- PC11.** Functional properties of high value end products.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Types of bio-waste suitable for vermicomposting.
- KU2.** Waste segregation and contamination risks.
- KU3.** Bio-waste collection, sorting, and pre-treatment.
- KU4.** Regulations around bio-waste use.
- KU5.** Safety and hygiene in waste handling.
- KU6.** Use of bins, shredders, and protective gear.
- KU7.** Scheduling waste input and turnover.
- KU8.** Organic vs. inorganic matter distinction.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Identify and sort compostable bio-waste.
- GS2.** Prepare organic matter for vermibeds.
- GS3.** Operate bio-waste processing tools.
- GS4.** Follow safety and sanitation protocols.
- GS5.** Track incoming waste volumes and types.
- GS6.** Communicate with suppliers of organic waste.
- GS7.** Manage odor and pest issues.
- GS8.** Coordinate disposal of unusable waste.
- GS9.** Record bio-waste data and source info.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Information about handling of waste</i>	6	-	-	-
PC1. Understand the composition of compost materials and vermimanure towards proper biowaste management.	2	-	-	-
PC2. Understand the difference between vermimanuring and vermiculture for handling biowaste materials.	2	-	-	-
PC3. Clearly understand the role of worms along with microbes used for conversion of Biobiowaste during vermimanuring.	2	-	-	-
<i>Knowledge on productivity and sustainable practice</i>	12	40	-	-
PC4. Understand detailed information about suitable biowastes for vermimanuring with proper providing parameters to obtain quality product for enhancing soil nutrition.	3	10	-	-
PC5. Knowledge on different sustainable processing practices for agriculture approaches.	3	10	-	-
PC6. Information about the different kinds of by-products and utilization in innovative agronomic process.	3	10	-	-
PC7. Know about best farming practices by integrating vermimanure for sustainable environmental approaches.	3	10	-	-
<i>Knowledge on production of high value end product</i>	12	30	-	-
PC8. Know the biochemistry and microbial aspects of different earthworm.	3	-	-	-
PC9. Selection and application of appropriate method for production of high value product.	3	10	-	-
PC10. Biochemical analysis of high value end products.	3	10	-	-
PC11. Functional properties of high value end products.	3	10	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0907
NOS Name	Operation And Biowaste Management
Sector	Agriculture
Sub-Sector	
Occupation	Animal Producers and Related Workers, Other
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024

Qualification Pack

CUTM/VSQ/N7803: Maintain Health and Safety at Work Place.

Description

This NOS focuses on ensuring a safe and healthy working environment by promoting hazard identification, safe equipment handling, proper use of PPE, and adherence to workplace safety protocols. It involves assessing and mitigating risks, responding to emergencies, and minimizing environmental and workplace hazards to protect self, co-workers, and surroundings.

Scope

The scope covers the following :

- The scope of this NOS includes performing regular safety checks, using safety gear, handling hazardous materials responsibly, following emergency procedures, and maintaining hygiene and cleanliness in the workplace. It aims to cultivate a culture of safety and environmental responsibility while reducing risk, injury, and workplace accidents.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Undertaking basic safety checks before operation of all machinery and vehicles and hazards are reported to the appropriate supervisor
- PC2.** Work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy.
- PC3.** Reading and understanding the hazards of use and contamination mentioned on the labels of pesticides/fumigants etc
- PC4.** Assessing the risks prior to performing manual handling jobs and work according to currently recommended safe practice.
- PC5.** Use equipment and materials safely and correctly and return the same to designated storage when not in use
- PC6.** Dispose of waste safely and correctly in a designated area
- PC7.** Recognize risks to bystanders and take action to reduce risk associated with jobs in the workplace
- PC8.** Perform your work in a manner, which minimizes environmental damage all procedures and work instructions for controlling risk are followed closely.
- PC9.** Report any accidents, incidents or problems immediately to an appropriate person and take necessary immediate action to reduce further danger.
- PC10.** Follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC11.** Follow emergency procedures to company standard / workplace requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Basic safety checks and pre-operational inspection procedures for machinery and vehicles.
- KU2.** Importance of reporting hazards to supervisors immediately.
- KU3.** Identification and correct use of personal protective equipment (PPE).
- KU4.** Hazard symbols, warning labels, and safe handling instructions on pesticides/fumigants.
- KU5.** Risk assessment techniques for manual handling and other workplace tasks.
- KU6.** Safe usage, handling, and storage of tools, materials, and equipment.
- KU7.** Proper methods of waste segregation and disposal.
- KU8.** Identifying workplace risks to others and preventive action planning.
- KU9.** Environmental protection procedures and pollution control practices.
- KU10.** Emergency response protocols including fire, accident, and evacuation procedures.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Perform routine safety inspections and hazard identification confidently.
- GS2.** Demonstrate use of PPE according to task-specific needs.
- GS3.** Interpret chemical hazard labels and apply safety practices accordingly.
- GS4.** Conduct basic risk assessments for manual work operations.
- GS5.** Operate tools and equipment safely, and store them properly after use.
- GS6.** Segregate and dispose of waste in compliance with workplace guidelines.
- GS7.** Monitor surroundings to identify risks to co-workers and intervene appropriately.
- GS8.** Follow environmentally responsible practices during work execution.
- GS9.** Report incidents promptly and initiate basic emergency response actions.
- GS10.** Respond effectively to accidents and emergencies as per protocol.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	30	-	-
PC1. Undertaking basic safety checks before operation of all machinery and vehicles and hazards are reported to the appropriate supervisor	-	-	-	-
PC2. Work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy.	2	3	-	-
PC3. Reading and understanding the hazards of use and contamination mentioned on the labels of pesticides/fumigants etc	2	3	-	-
PC4. Assessing the risks prior to performing manual handling jobs and work according to currently recommended safe practice.	2	3	-	-
PC5. Use equipment and materials safely and correctly and return the same to designated storage when not in use	2	3	-	-
PC6. Dispose of waste safely and correctly in a designated area	2	3	-	-
PC7. Recognize risks to bystanders and take action to reduce risk associated with jobs in the workplace	2	3	-	-
PC8. Perform your work in a manner, which minimizes environmental damage all procedures and work instructions for controlling risk are followed closely.	2	3	-	-
PC9. Report any accidents, incidents or problems immediately to an appropriate person and take necessary immediate action to reduce further danger.	2	3	-	-
PC10. Follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Follow emergency procedures to company standard / workplace requirements	2	3	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7803
NOS Name	Maintain Health and Safety at Work Place.
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services



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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.



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3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the AB
- Questions are mapped to the specified assessment criteria.
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location.
- Centre photographs with signboards and scheme specific branding
- Video Call verification/online proctoring will be done during the assessment from AB.

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Minimum Percentage Required

- 70% overall to pass.
- Minimum 70% in each NOS

7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

8. Assessor Details: ToA Certified Assessor with relevant experience.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 60

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

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Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N0901.Handling of Vermimanuring	30	40	-	30	100	10
CUTM/AGR/N0902.The Role of Earthworms in Vermimanuring	20	30	-	-	50	5
CUTM/AGR/N0903.Choosing the Right Earthworm Species	20	30	-	-	50	5
CUTM/AGR/N0904.Advanced Vermimanuring Techniques	30	70	-	-	100	10
CUTM/AGR/N0905.Value added to Vermimanuring and Preservation	40	60	-	-	100	10
CUTM/AGR/N0906.Quality Assurance in Vermimanuring	40	60	-	-	100	10
CUTM/AGR/N0907.Operation And Biowaste Management	30	70	-	-	100	20
CUTM/VSQ/N7803.Maintain Health and Safety at Work Place.	20	30	-	-	50	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	20
Total	250	420	-	30	700	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.